

Multi-criteria energy system analysis of onshore wind power distribution in climate-neutral Germany

C. Lohr^a, F. Peterssen^c, M. Schlemminger^b, A. Bensmann^{a,*}, R. Niepelt^{b,c}, R. Brendel^{b,c}, R. Hanke-Rauschenbach^a

^a Leibniz University Hannover, Institute of Electric Power Systems, Appelstr. 9A, Hannover 30167, Germany

^b Leibniz University Hannover, Institute of Solid State Physics, Appelstr. 2, Hannover 30167, Germany

^c Institute for Solar Energy Research Hamelin (ISFH), Am Ohrberg 1, Emmerthal 31860, Germany

ARTICLE INFO

Keywords:

energy system analysis
wind power
spatial distribution
system integration
multi-criteria-optimization
energy system modeling

ABSTRACT

Although onshore wind energy is a key pillar of renewable energy systems, installation targets in Europe have not been met. One contentious issue is its distribution, involving trade-offs between economic costs, environmental impact, public acceptance, and equity considerations. In this study, we evaluate different distribution strategies that meet Germany's national onshore wind power target of utilizing 2 % land area, breaking it down to subordinate levels such as federal states. Therefore, we define key indicators for energy policy objectives to comprehensively analyze these strategies. We employ an energy system optimization model to address the system integration of spatial onshore wind power distribution, an aspect often overlooked in previous studies. Our results indicate that the impact of different distribution strategies on the overall energy system design is moderate, with the highest sensitivity observed in the allocation of electrolyzers, which closely align with renewable energy surpluses. However, our analysis shows that concentrating onshore wind power in areas with high energy yield can lead to an increase in electricity transport by up to 38 %, whereas more evenly distributed scenarios are preferred for environmental sustainability and distributive justice. In conclusion, we argue that energy system analysis can enhance the accuracy of assessment of onshore wind power distribution, but it must consider non-techno-economic criteria within spatially-distributed energy systems itself to address policymakers.

1. Introduction

The subject of the present article is the analysis and evaluation of distribution strategies for onshore wind power, which is a key pillar in carbon-neutral energy systems (IEA, 2021). Despite the globally increasing installations of onshore wind farms, the current growth rates have not been sufficient to comply with limiting global warming to well below 2 °C (GWEC, 2022). This is particularly true for Europe, where new installations have been stagnant, and targets are expected to continue to be missed clearly until 2026 (WindEurope, 2022). Several factors contribute to this slow growth, including lengthy project approval procedures, a lack of contracts for difference, slow grid extension, and the lack of designated priority areas for installing onshore

wind power (WindEurope, 2022; Reutter et al., 2022). To address these issues, it is necessary to break down national installation targets into subordinate levels to coordinate new projects and also overcome the free-rider problem of multi-level governance (Reutter et al., 2022). We propose three key steps for establishing a distribution strategy, which various types of literature contribute to: assessing the wind energy potential (1), defining distribution criteria (2), and applying a distribution method (3).

Firstly, detailed information on wind energy potential needs to be provided. Geographical, technical, economic, and feasible wind energy potential definitions need to be differentiated (McKenna et al., 2022). This involves determining the eligible land area, typically handled reversely based on exclusion criteria (geographical potential), as well as

Abbreviations¹²: CRV, Conflict risk value; DC, Direct current; ESTRAM, Energy System Transformation Model; G, Giga 10⁹; K, Kilo 10³; LCOE, Levelized cost of electricity; M, Mega 10⁶; NUTS, Nomenclature of territorial units for statistics; P, Peta 10¹²; SR, Space resistance.

* Corresponding author.

E-mail addresses: lohr@ifes.uni-hannover.de (C. Lohr), peterssen@isfh.de (F. Peterssen), schlemminger@solar.uni-hannover.de (M. Schlemminger), astrid.bensmann@ifes.uni-hannover.de (A. Bensmann), niepelt@solar.uni-hannover.de (R. Niepelt), brendel@solar.uni-hannover.de (R. Brendel), hanke-rauschenbach@ifes.uni-hannover.de (R. Hanke-Rauschenbach).

<https://doi.org/10.1016/j.egy.2024.07.064>

Received 26 June 2023; Received in revised form 28 July 2024; Accepted 30 July 2024

Available online 10 August 2024

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meteorological data and turbine technologies, to assess energy yield (technical potential). Afterward, further constraints such as economic (economic potential) and non-economic aspects (feasible potential) can be applied. In addition to hard-constrained eligible land areas, qualitative information, such as conflict risk values (Riedl et al., 2020; Stemmer et al., 2021) or suitability factors (McKenna et al., 2014; Harper et al., 2019) can be included. This allows prioritization and consideration of a feasible potential for the necessity of reaching a set target. Definitions, data, and criteria on wind energy potential vary across literature (McKenna et al., 2022) resulting in a wide range of technical potential for Europe (McKenna et al., 2015; Ryberg et al., 2019; Ruiz et al., 2019).

Secondly, criteria need to be defined to answer how to distribute a set target to subordinate levels. Distribution criteria often represent conflicting objectives. Historic drivers for onshore wind power siting in Europe have prioritized cost-effectiveness under consideration of low local disamenities (Weinand et al., 2022a). For future expansion in highly renewable energy systems, Lehmann et al. (2021) suggested wind yield, load proximity, natural and landscape conservation, and interregional equity as criteria and intended to rank them. However, they concluded that *one* optimal distribution cannot exist, as depending on the stakeholder group not only the rankings but also criteria definitions differed. Other criteria and indicators have been suggested in the literature that involved bottom-up modeling with a focus on spatially-explicit turbine placing. Those include the calculation of a turbine performance index based on energy yield, bird collision risk, and settlement distance (Eichhorn et al., 2019), or single criteria trade-off analysis between levelized cost of electricity (LCOE), network length, loss of scenicness, and distribution equality (Weinand et al., 2022b), or between generation costs, disamenity costs for residents (Grimsrud et al., 2021; Ruhnau et al., 2022; Lehmann et al., 2023), aesthetic landscape quality, and nature conservation (Tafarte and Lehmann, 2023).

Thirdly, in practice, a distribution method has to be applied that considers more than one criterion pursuing multiple objectives at the same time. Drechsler et al. (2011) used the willingness-to-pay method to define weights for their welfare optimization, which includes production costs and externalities. Due to existing administrative borders, other examples focused on national distributions (top-down). Several studies broke down the national land area target of 2 % (Bofinger et al., 2012; SPD et al., 2021) for Germany to the federal state level. Stemmer et al. (2021), first, defined a realistic potential per state and, then, utilized it proportionally to reach the set national land area goal. In *Stiftung Klimaneutralität* (2021), a similar method was used but, additionally, weights regarding varying wind resources were included. Guidehouse et al. (2022) applied an iterative process that provided equal results around 2 % land area utilization for all states that were marginally adjusted based on a state's relative available potential.

In Table 1, we summarize the selected examples from the literature highlighting their contributed indicators categorized into the (partly overlapping) distribution criteria of economic costs, environmental degradation, and equity and public acceptance. In literature with spatially-explicit turbine placement, trade-offs between single indicators have been shown with detailed bottom-up models (Weinand et al., 2022b; Tafarte and Lehmann, 2023). However, in the presented literature, in which capacity targets have been broken down, much simpler methods beginning from an equal distribution have been applied, sometimes ignoring even wind conditions. This can be explained to a certain extent because valid indicators do not exhaustively cover an objective (e.g. bird collision risks cannot solely represent environmental degradation) and selecting and weighting criteria remain contestable. Furthermore, isolated indicators, such as LCOE or the related energy yield, have been used to approximate the economic value of onshore

wind power plants. However, these do not consider the integration into the energy system, which may skew the economic assessment and has been identified as a research need in previous studies (McKenna et al., 2022; Weinand et al., 2022a, 2022b; Lehmann et al., 2023; Tafarte and Lehmann, 2023).

Traditional research on onshore wind power distribution has not fully considered the broader energy system, while energy system modeling has often prioritized system costs, neglecting environmental and social aspects (Süsser et al., 2022; Vågerö and Zeyringer, 2023). There are a few notable exceptions in the literature. Grimsrud et al. (2021) consider broadly defined environmental costs in the allocation of wind turbines in Norway, and Price et al. (2022) incorporate scenic landscapes in the optimization of Great Britain's power system.

In this study, we apply an energy system optimization model to distribute onshore wind power capacity among the federal states in a climate-neutral Germany. Our model approach features an aggregated representation of environmental degradation and includes equality aspects. In contrast to the existing top-down approaches in the literature, we use an energy system model and link onshore wind power with the supply of other renewable sources to meet a given hourly energy demand. As we also consider limited transport capacities between the federal states and the need for energy storage and sector-coupling, we improve the economic assessment of onshore wind power by accounting for different costs of system integration. Besides ensuring technical feasibility, applying an energy system model allows the quantification of systemic consequences of different distributions. In this paper, we suggest a multi-criteria framework to compare three different distribution strategies and assess their impact on relevant energy supply objectives.

Thereby, we pursue two main research questions:

- How do different onshore wind power distribution strategies impact the energy system and key objectives of the energy supply in a climate-neutral Germany?
- What insights can energy system modeling provide for the distribution of onshore wind power in general?

Our study makes a methodological contribution by integrating the fields of onshore wind power distribution and energy system analysis and providing a comprehensive analysis of various distribution strategies and energy policy objectives. Furthermore, the quantified trade-offs presented in our results are relevant for policymakers, offering insights into the impacts of different distribution strategies on key objectives.

The article is structured as follows: in Section 2, we give an overview of the energy system optimization model and the data applied in this study. Section 3 contains descriptive results of the onshore wind power distribution and their corresponding energy system designs as well as an impact analysis on relevant objectives of energy supply. Section 4 discusses our results regarding the outlined research questions, from which we conclude in Section 5.

2. Methods

2.1. Model setup

The linear optimization model applied in this article is based on the ESTRAM framework (Lohr et al., 2022) and solved with Gurobi (Gurobi Optimization LL, 2021). For a scenario of a climate-neutral Germany in the year 2045 (KSG, 2021), we minimize the costs for the system integration of renewable energy to cover a projected hourly energy demand. We use the same set of equations described in Lohr et al. (2022), which is amended by further features explained in this section.

We predefine a limited solution space within our model, which is based on the results of an existing study (cf. Section 2.2), to focus on the distribution of onshore wind power and keep the results of different scenarios comparable. Consequently, the total installed capacity (sum over all regions) for all renewable energy sources is given exogenously.

¹ Abbreviations of federal states are listed in Table 2. The nomenclature for the model is collected in Tables A9–A11 in the appendix

Table 1

Wind energy distribution approaches and mentioned indicators of distribution criteria in selected literature.

Objective	Studies	Potential data (1)	Distribution criteria (2)			Distribution method (3)
			Economic costs	Environmental degradation	Equity and public acceptance	
Criteria weights	Lehmann et al. (2021)	Masurowski (2016)	Energy yield, load proximity*	Nature and landscape conservation risks	Grid expansion*, Interregional equity	Participatory stakeholder involvement
	Drechsler et al. (2011)	Own approach	Energy yield, production costs	Bird and bat population loss	Minimum settlement distance	Optimization
Spatially-explicit turbine placement (bottom-up)	Eichhorn et al. (2019)	Own approach	Energy yield	Bird collision	Settlement distance	Optimization
	Weinand et al. (2022b)	Ryberg et al. (2019)	LCOE, network length	Loss of scenicness	Equality of turbines per inhabitant	Optimization
	Tafarte, Lehmann (2023)	Masurowski (2016)	Generation costs	Nature conservation and aesthetic landscape quality impact	Disamenity costs	Optimization
	Stiftung Klimaneutralität (2021)	Own approach	Wind resources		Equality of capacity potential utilization	Proportional
Capacity distribution (top-down)	Stemmer et al. (2021)	Based on Riedl et al. (2020)		Exclusion of areas with (high) conflict risks	Equality of capacity potential utilization	Proportional
	Guidehouse et al. (2022)	Own approach		Conflict risks	Equality of capacity potential and land area utilization	Iterative process
	This study	Guidehouse et al. (2022)	Energy yield, system integration	Space resistance of vulnerable area	Equally distributed** area vulnerability (Lohr et al., 2022)	Optimization

* load proximity and grid expansion overlap

** This modeling approach approximates that every region has capacity potential equally distributed such as from far away to close to settlement. Interregional differences between regions (e.g. in biodiversity or scenicness) are not accounted for.

Except for onshore wind power plants, this includes also their distribution on the considered model nodes,² which represent the federal

Table 2

Lists of the considered regions and their abbreviations.

NUTS-code	Name	Abbreviation	Land area in km ²	Currently installed onshore wind power capacity in GW ^a
DE1	Baden-Württemberg	BW	35748	2
DE2	Bavaria	BY	70542	3
DE3	Berlin	B	891	0.02
DE4	Brandenburg	BB	29654	9
DE5	Bremen	HB	420	0.2
DE6	Hamburg	HH	755	0.1
DE7	Hesse	HE	21116	3
DE8	Mecklenburg-Vorpommern	MV	23295	4
DE9	Lower Saxony	NI	47710	13
DEA	North Rhine-Westphalia	NW	34112	7
DEB	Rhineland-Palatinate	RP	19858	4
DEC	Saarland	SL	2571	1
DED	Saxony	SN	18450	1
DEE	Saxony-Anhalt	ST	20459	5
DEF	Schleswig-Holstein	SH	15804	9
DEG	Thuringia	TH	16202	2
DE	Germany	DE	357588	61

^a Based on Deutsche WindGuard GmbH (2024)

² It is generally desirable to co-optimize the distribution of other renewable technologies, in particular solar energy. However, given that the cost-optimal distribution of solar energy may not be a feasible or probable solution, we rely on exogenous assumptions.

states (NUTS1-level) of Germany (cf. Table 2). Other decision variables of the optimization model are the quantity and allocation of capacities of converters and storages, as well as their dispatch, and the energy carrier import. Those variables account for the different needs for flexibility of the onshore wind power distributions and keep the model feasible. Additionally, the import of hydrogen (H₂) serves as a slack to balance the varying onshore wind power yield. For the energy carrier biomass, we set an upper energetic limit, which reflects the aggregated sustainable domestic potential of heterogeneous compounds such as waste, residues, and energy crops. The energy carriers electricity, H₂, and gas (methane) can be transported via grids. All considered components and energy conversions are depicted in Fig. 1. Since our focus is solely on Germany, we have omitted potential synergies of the European energy system, such as continental power balancing.

The objective function comprises the costs for the capacities $P_{c,k}^{cap}$ of converters and storages summed over all nodes k , as well as the energy carrier costs $c_{H_2,k}^{car}$ for the imported H₂ energy $E_{H_2,k}^{import}$, which together, we label as the system integration costs³ (cf. Eq. 1). The costs for the considered components c are annualized and include the capital expenditures (derived from specific fix costs c_c^{fix} and the capital recovery factor β_c^{crf}) and operation and maintenance c_c^{om} .

$$C^{sys} = \sum_k \left(E_{H_2,k}^{import} \cdot c_{H_2,k}^{car} + \sum_c P_{c,k}^{cap} \cdot c_c^{fix} \cdot (\beta_c^{crf} + c_c^{om}) \right) \quad (1)$$

The cost term cannot be attributed to onshore wind power solely and unlike other definitions also comprises costs for other primary energy such as H₂. However, the differences in the absolute values can serve as opportunity costs between different onshore wind power distributions as other renewable sources remain unchanged. For selected scenarios, the system integration costs are complemented by quadratically increasing environmental costs, representing a negative impact due to the land

³ The deterministic model approach considers profiling and network costs but does not account for balancing costs (Hirth et al., 2015).

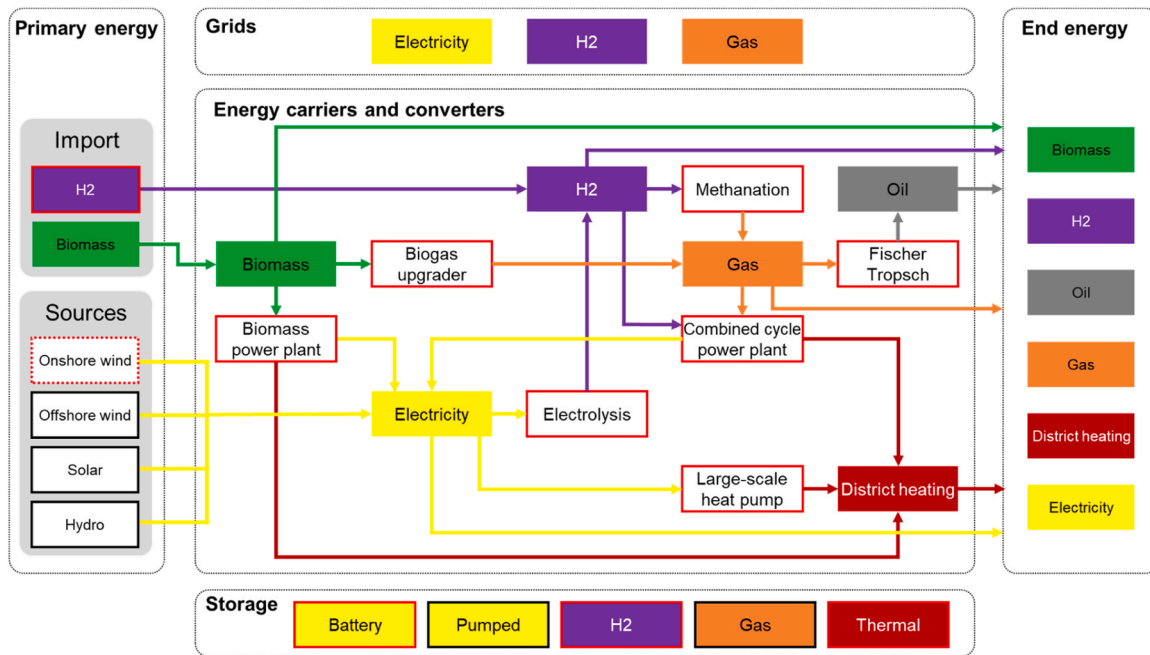


Fig. 1. Energy system topology of the applied model representing a single node. The capacity of components with black frames is set to a fixed value, whereas red frames signalize optimization. For onshore wind power, the total capacity over all nodes remains constant as well, however, the spatial distribution varies.

consumption of onshore wind power plants (Lohr et al., 2022).

To calculate the environmental costs C^{env} , we use a reference land area consumption by onshore wind power capacity, which is multiplied by the maximum space resistance (SR) $c^{\text{sr,max}}$ as a cost coefficient that also determines the weight within the objective function (cf. Eq. 2). The reference land area consumption is computed by the product of onshore wind power capacity $P_{\text{OW},k}^{\text{cap}}$ and the utilization of the capacity potential $\frac{p_{\text{OW},k}^{\text{cap}}}{p_{\text{OW},k}^{\text{max}}}$ over all nodes k , as well as the capacity density ϑ of onshore wind power. This means that the first installation in a state only marginally contributes to the reference land area consumption, whereas the last consumed square kilometer adds 1 km^2 . Therefore, the resulting quadratic term reaches its maximum if all onshore wind power capacity is allocated at nodes with full capacity. This takes into account that if the capacity potential is utilized only partially, environmental criteria can be considered when placing the wind turbines. (Lohr et al., 2022)

$$C^{\text{env}} = c^{\text{sr,max}} \cdot \sum_k \frac{P_{\text{OW},k}^{\text{cap}}}{2 \cdot \vartheta \cdot p_{\text{OW},k}^{\text{max}}} \quad (2)$$

Furthermore, new equations have been added to the model (cf. Appendix Eq. A3-7). For the alternating current branches within the electricity grid, we apply a direct current (DC) power flow model (Hörsch et al., 2018). The combined cycle power plant allows the dual input of gas and H_2 . Similar to the biomass power plant, it is modeled as a combined heat and power plant with a flexible heat-to-power output ratio (Kunz et al., 2017).

2.2. Data

We use the data of annual end energy demand and renewable energy based on a study conducted by Prognos et al. (2021) and commissioned by Agora Energiewende to create an energy system for all our scenarios. The study outlines transition pathways for a climate-neutral energy system in Germany by 2045 but lacks a spatial dimension. Therefore, we apply our own methods to create individual time series per region. For the spatial distribution, we map the given information with data on the present sectoral energy consumption per state (LAK Energiebilanzen, 2022). The aggregated end energy per carrier type and region is

collected in Table A12 in the appendix. For the temporal distribution, we generate time series per sector and aggregate them. Therefore, we use different methods to receive the time series of households (Schlemminger et al., 2021), mobility (De Tena, 2014), and the tertiary sector (Dolman et al., 2012). The industrial demand, we approximate with a periodic step-profile that considers less activity on weekends (60 %) and at night (80 %).

For renewable energy, we use data on the total installed capacity of onshore wind (145 GW), offshore wind (70 GW), and solar power plants (385 GW), as well as a biomass potential of 347 TWh (Prognos et al., 2021). As hydro power plant capacity, we apply the currently installed 4 GW (Anderer et al., 2012). The spatial distribution of solar power plants is subdivided into large-scale and rooftop plants. For solar large-scale, we assume full utilization of the capacity potential (230 GW) given by Kelm et al. (2019), whereas the remaining solar rooftop capacity of 155 GW and offshore wind capacity are extrapolated based on existing installations (Bundesnetzagentur, 2019). The distribution of onshore wind power capacity is the subject matter of this article and remains variable. To generate capacity factors as time series, we use the coastdat-2 COSMO-CLM reanalysis (Geyer and Rockel, 2013) as weather data and apply it to reference plant models (Holmgren et al., 2018; Haas et al., 2021). For onshore wind power, we use the Enercon E 92 2300 for average wind speeds above 6 m/s or otherwise the Enercon E 115 2500 as reference plants. The resulting maximum total electricity generation (without curtailment) of the considered technologies per region is shown in Table A13 in the appendix.

An overview of the onshore wind power capacity potential and the maximum full load hours per state is shown in Table 3. For the targeted 145 GW onshore wind power capacity, we assume a total land area coverage of 2 % as a reference. The implied capacity density of 20 MW/ km^2 is slightly below values in the studies of Masurowski (2016) (21.5 MW/ km^2) and Reiner Lemoine Institut (2022) (21 MW/ km^2) but above other examples in McKenna et al. (2022). It is important to note that there exists a significant discrepancy between studies regarding the estimated onshore wind power capacity potential, which can influence the quantitative results of our analysis. We calculate the capacity potential based on the results of the study by Guidehouse et al. (2022) commissioned by the Federal Ministry for Economic Affairs and Climate

Table 3

Onshore wind power capacity potential and maximum full load hours per federal state of this study in comparison to selected literature.

	Capacity potential* in GW				Maximum full load hours		
	This study	Masuiowski (2016)	Stemmer et al. (2021)	Reiner Lemoine Institut, (2022)	This study	Masuiowski (2016)	Luderer et al. (2021)
BW	19	20	10	27	1896	1895	2631
BY	46	49	22	39	1966	1813	2546
B	0.05	0	0.06	0.03	2546	-	-
BB	54	33	37	34	2414	2964	3666
HB	0.04	0	0.04	0.12	3042	-	-
HH	0.04	0	0.04	0.09	2465	3144	4096
HE	25	8	7	16	2318	2579	2934
MV	20	80	19	26	2571	3544	3595
NI	47	39	58	94	2808	3604	3616
NW	17	8	13	34	2458	3103	3560
RP	17	9	7	13	2264	2186	2804
SL	1.5	0.2	0.12	0.7	2310	2180	2857
SN	14	22	10	8	2508	3107	3350
ST	27	64	44	47	2355	3013	3371
SH	12	15	12	34	3457	3735	3732
TH	23	22	17	23	2505	2786	2995
DE	323	369	256	395	2418	2832	3254

* a capacity density of 20 MW/km² is assumed if only data on the land area is available.

Action of Germany. The authors provide information about land area-specific conflict risks due to the erection of onshore wind turbines but suggest that in total 4.5 % of the national land area is eligible for onshore wind power.

The capacity factors are calculated with data from the weather year 2012. Additionally, we assume a yield deviation of +/-10 % from the average capacity factors for every state to account for intra-regional differences and decreasing yield due to wake effects (Lohr et al., 2022). The resulting 2418 maximum full load hours for Germany comprise the average for the total capacity potential and, therefore, in practice, can be interpreted as a lower bound. The estimated energy yield is higher than in Prognos et al. (2021) (2200 h), however, is exceeded clearly by values from Masuiowski (2016) and Luderer et al. (2021).

Other data comprise storage, imports, and grid infrastructure as well as costs and technological parameters. Existing gas storage and pumped hydro storage facilities can be used and remain available without new investment (Prognos et al., 2021), whereas battery, H₂, and thermal storage power and energy capacity are optimized. Besides the utilization of the energy carrier biomass, only an import of H₂ is represented. The H₂ import is unconstrained in quantity, set constant in time to prevent unrealistic flexibility potential, and available at costs of 110 EUR/MWh (3.67 EUR/kg). Other synthetic climate-neutral energy carriers as import options are not available in the model but synthetic gas and oil can be obtained domestically as H₂ derivatives. Carbon capture and storage or utilization are not considered.

Data of the present electricity grid for 110 kV, 220 kV, and 330 kV (Mueller et al., 2018) is supplemented by already planned expansion projects (Bundesnetzagentur, 2020) but is limited to 70 % of each line capacity to account for security margins (Deutsche Energie-Agentur GmbH, 2010). Furthermore, the existing gas infrastructure (Pluta et al., 2022) is equally split for synthetic gas and H₂ transport. For all energy carriers with grid infrastructure, transport capacities between regions are aggregated, whereas transport losses depend on the distance between centroids and linearized specific losses (cf. Table 4).

Costs and other parameters of technologies can be found in the appendix Table A14.

Table 4

Specific transport losses per distance and energy carrier.

Energy carrier	Specific transport losses /100 km
Electricity	1.3 % (Oeding and Oswald, 2016)
H ₂	0.8 % (Merten et al., 2020)
Gas	0.2 % (Gö8, 2017)

2.3. Energy system evaluation criteria

As onshore wind power is part of the energy system, we assess the distribution strategies with energy system evaluation criteria⁴ in this study. Therefore, we suggest evaluation criteria based on the three goals of energy policy derived from German § 1 EnWG (2022). For the **affordability** of energy, we look at the differences in system integration costs mentioned in Section 2.1. The goal of energy supply security is divided into two criteria: political **energy independence** and **system reliability** within the country. The latter criterion represents the technical perspective of supply security. As all scenarios are already defined as climate-neutral without CO₂ emissions, **environmental sustainability** includes the negative impact due to land consumption by onshore wind power plants. Additionally, we incorporate **distributive justice** as a criterion to include a different dimension of public acceptance, which is a necessary precondition in democratic countries (Local Energy Consulting, 2020). Therefore, we measure the regional equality of the onshore wind power distribution, which follows previous research in the fields of onshore wind power distribution (Weinand et al., 2022b) and energy system analysis (Sasse and Trutnevyte, 2020). For each criterion, we use different key indicators, which are explained in detail in Section 3.3.

3. Results

This section provides an overview of the onshore wind power distribution scenarios and their corresponding energy system designs. Furthermore, the impact of the different scenarios on the energy system evaluation criteria provided in Section 2.3 is analyzed.

3.1. Scenario definitions

We analyze three different distribution strategies that are represented by our scenarios illustrated in Table 5. We use the proposed 2 % of land area for onshore wind energy as a reference (SPD et al., 2021), which equals an installed capacity of 145 GW in this study. Therefore, we define an *Equal2%* scenario, in which 2 % of the land area is utilized in all 16 federal states of Germany. This scenario serves as a reference, in which the onshore wind distribution is fixed. To meet the energy demand at any time and location, the system flexibility needs to be

⁴ The evaluation criteria that assess the energy system need to be distinguished from the onshore wind power distribution criteria presented in Table 1.

Table 5

Definition of onshore wind power distribution scenarios including differences in decision variables and optimization criteria (part of the objective function).

	Onshore wind power distribution scenarios		
	Equal2%	Cost-optimal	Socio-cost-optimal
Onshore wind power distribution	fixed	variable	variable
Dispatch	variable	variable	variable
System integration*	optimized	optimized	optimized
Environmental impact	-	-	optimized**

* System integration comprises costs for converters, storages, and H₂ import

** Variation of maximum space resistance parameter for socio-cost-optimal scenarios

provided by the optimized capacity of H₂ import, storage, and converters.

In contrast to the *Equal2%* scenario, in the *Cost-optimal* scenario, the onshore wind power capacity distribution is variable within the constraining bounds of capacity potential described in 2.2. This way, lower energy system costs can be achieved due to a more efficient allocation of onshore wind power capacities at sites with higher yields.

At last, the *Socio-cost-optimal* scenarios include a multi-criteria optimization, in which environmental costs are considered (Lohr et al., 2022). As both, weighting and quantification of environmental impact are capious, we vary its cost coefficient, the maximum SR parameter, up to 10 M EUR/km², and provide further eight scenarios to circumvent subjective outcomes.⁵ At the lower end, the resulting onshore wind power distributions equal the *Cost-optimal* scenario. Throughout all scenarios, the dispatch of all components and the transport flows are variable.

3.2. Onshore wind power distribution

Fig. 2 depicts the resulting optimized onshore wind power distributions of all scenarios using a) installed capacity, b) land consumption, and c) capacity potential utilization⁶ as metrics. Each scenario's solution includes 145 GW of installed onshore wind power capacity, corresponding to the national goal of utilizing 2 % of the total land area in Germany.

The *Equal2%* scenario with its fixed onshore wind power capacities exceeds the given capacity potential in the three smallest states.⁷ On the other hand, the capacity potential utilization in four larger states (Brandenburg, Hesse, Saxony-Anhalt, and Thuringia) is below one third. The *Cost-optimal* distribution is influenced by the energy yield and therefore, the scenario retains a concentration of wind power in the north and center of Germany. In total, half of the states utilize the full capacity potential, whereas six states have little or no onshore wind power capacity installations at all. This includes Brandenburg and Bavaria, which, besides Lower Saxony, hold the largest eligible land areas of all states. An increasing maximum SR shifts the *Socio-cost-optimal* scenarios toward a capacity potential utilization of 45 % in all states. Consequently, the installations, and thus the resulting land consumption, are sensitive to the given capacity potential. Regardless, for a minimum parameter value of 1 M EUR/km², onshore wind power capacity is erected in every state.

A difference in onshore wind power distribution also causes a

⁵ at a site with 2500 full load hours this implies maximum marginal (environmental) costs of 200 EUR/MWh for the highest value of the space resistance parameter if the full capacity potential is utilized.

⁶ Capital potential utilization refers to the extent to which the available capacity for onshore wind power is used in each region

⁷ To maintain the technical feasibility of the model, we removed the capacity potential constraint for the *Equal2%* scenario due to the fixed distribution.

relocation of other components, such as converters and storages. E.g. electrolysis distribution, shown in Fig. 3, has a strong positive correlation to onshore wind power capacities (maximum correlation of 0.87 in the *Socio-cost-optimal* scenario with a maximum SR of 1 M EUR/km²), which, in general, contributes to an electricity surplus. Correspondingly, offshore wind energy in the north causes a high plateau of electrolysis in the pertaining states and the focus of H₂ production in the north. In terms of absolute quantity, higher yield and electricity surplus in the *Cost-optimal* scenario lead to 9 % more capacity of electrolysis compared to *Equal2%*.

Storage components do not show a significant correlation with wind energy capacity distribution and remain mostly unaffected. Thereof, optimized battery storage capacities vary the most (671–690 GWh) but are primarily determined by solar energy sites. Besides, the fixed capacity of hydro power plants (constant 4.3 GW) and pump storage (5.9 GW), other components' capacities and distribution are also not significantly affected by wind energy distribution as indicated for selected technologies in Table 6.

3.3. Impact analysis

In this section, we assess the onshore wind power distribution scenarios using the energy system evaluation criteria described in Section 2.3. For each criterion, we have defined key indicators, which are explained and applied in the following. We compare the three distribution scenarios: *Equal2%*, *Cost-optimal*, and *Socio-cost-optimal SR10*. As the *Socio-cost-optimal* scenario, we specifically selected the extreme scenario (highest value for maximum SR), in which the onshore wind power capacity potential utilization of all regions converges.

3.3.1. Affordability

To indicate the differences in energy affordability within the scenarios, we compare the (annual) system integration costs. These comprise capital expenditures, operation and maintenance costs of the optimized converters and storages as well as the energy carrier costs for H₂ import. Other parts of system costs, such as for (other) primary energy (41.7 G EUR), are identical in all scenarios and subsequently neglected.

The general cost structure, illustrated in Fig. 4, does not depend significantly on the onshore wind energy distribution and primarily consists of H₂ import costs. The *Cost-optimal* scenario reduces the system integration costs by 2.4 G EUR annually compared to the *Equal2%* scenario. The efficient onshore wind energy allocation increases electricity generation and, thus, primary energy, which otherwise is substituted by H₂ imports. Whereas this leads to savings of 2.7 G EUR, increased capacities of electrolysis and battery storage (82 % of all storage costs) slightly reduce the cost advantage. The *Socio-cost-optimal SR10* distribution is in between the other scenarios and retains only 0.6 G EUR of savings.

The total cost differences appear marginal, however, the savings account for up to 23 % of the expected annual asset value of onshore wind power plants⁸ (11.7 G EUR). Shadow prices show that an average 3 MW turbine in fully utilized Schleswig-Holstein with annual costs of 241 k EUR could reduce the system costs by 219 k EUR in the *Cost-optimal* scenario, while other sites remain unused. This implies a big gap in individual turbine profitability, even more, pronounced in the other scenarios.

3.3.2. Energy independence

Energy independence reflects the political dimension of energy security, which we measure with the need for H₂ imports. All scenarios have the same total installed onshore wind power capacity of 145 GW, however, the yield varies between the regions and, thus, the provided

⁸ Contains capital costs and operation and maintenance costs (cf. Table A14)

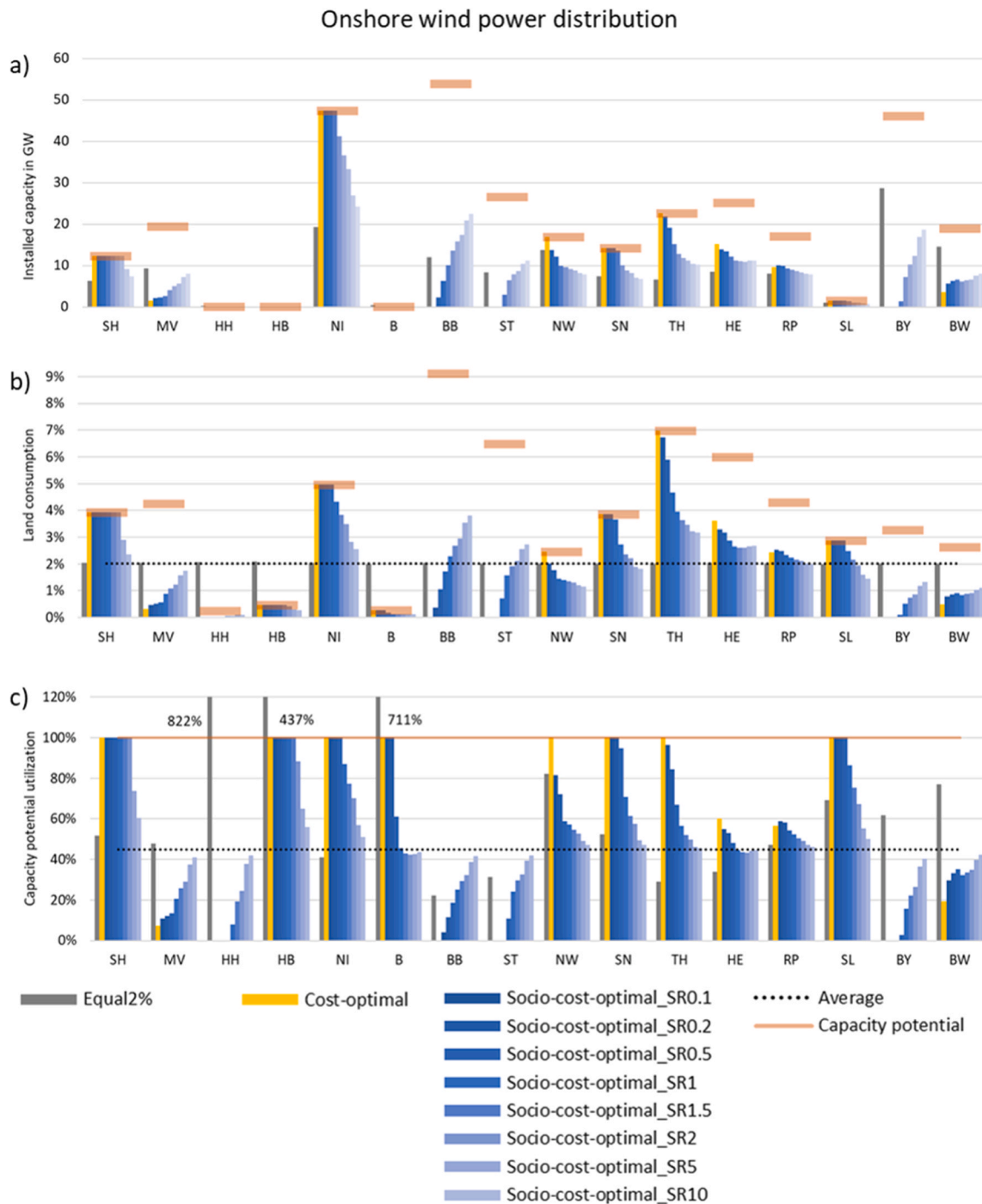


Fig. 2. Resulting optimized onshore wind power capacity distribution as a) installed capacity, b) land consumption, and c) capacity potential utilization per scenario and federal state sorted from North to South.

onshore wind energy. Therefore, Fig. 5 contrasts the H₂ imports with the onshore wind energy supply. The additional energy from other sources does not vary more than 1 TWh. This comprises 364 TWh of solar energy, 249 TWh of offshore wind energy, 16 TWh of hydro energy, and 347 TWh of additional primary bioenergy.

Due to the more efficient allocation, 11 % more onshore wind energy is supplied in the *Cost-optimal* scenario compared to the *Equal2%* scenario. This means 5 % fewer H₂ imports and, therefore, higher energy independence. The surplus of 37 TWh of electricity, however, can only save 26 TWh of H₂ imports as the lossy local production by electrolysis is higher by 20 TWh (185 TWh in total). This leaves the *Equal2%* scenario

with the lowest total primary energy supply (1804 TWh). The H₂ import, as well as the wind energy and primary energy supply, within the *Socio-cost-optimal_SR10* scenario, lies between the other two scenarios.

3.3.3. System reliability

Even though all energy flows satisfy the given model constraints, we evaluate the technical energy supply security of the different scenarios and account for model limitations. These include not only the coarse spatial resolution with the associated aggregation of line capacities and neglect of local bottlenecks but also the disregard of transit flows in the European transmission systems. Therefore, we calculate the energy

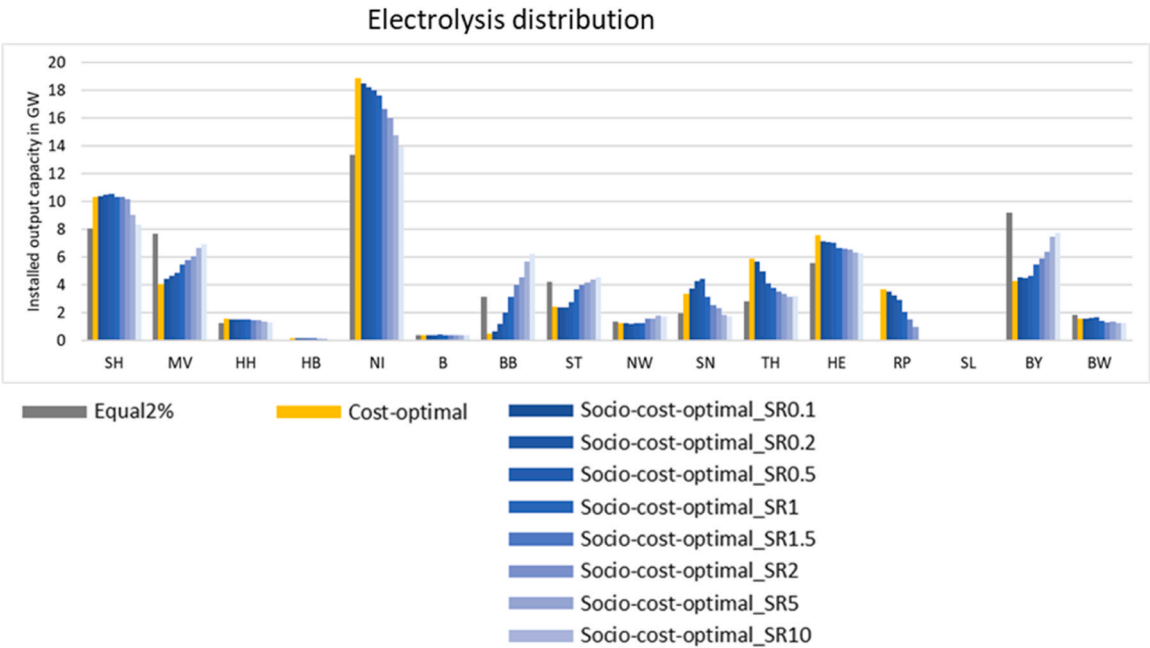


Fig. 3. Electrolysis capacity distribution as installed capacity (H₂) per scenario and federal state sorted from North to South.

Table 6
Non-renewable electricity generation and consumption output capacity per scenario summed over all federal states.

Technologies	Total installed capacity in GW _{out}		
	Equal2%	Cost-optimal	Socio-cost-optimal_SR10
Biomass power plant	1.5	1.2	1.5
Combined cycle power plant	50.6	52.5	52.4
Battery storage	135.0	135.5	134.1
Electrolysis	61.0	66.2	63.9
Large-scale heat pump	45.6	45.0	45.0

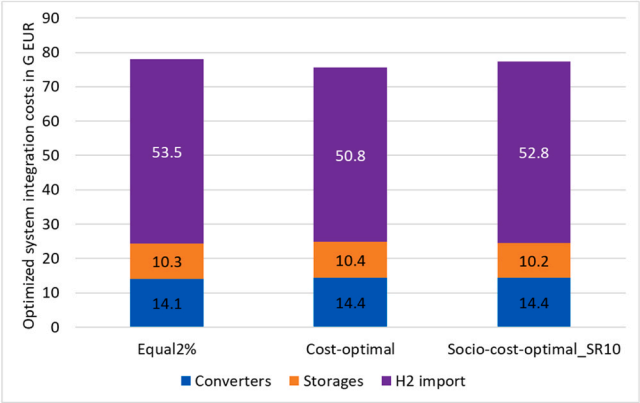


Fig. 4. Optimized system integration costs split into converters, storages, and H₂ import per scenario.

transport as the accumulated energy flows multiplied by the distance between nodes for the energy carriers with grid infrastructure. We use energy transport as an indicator of the risk of either potential threats to system reliability or an increased need for grid expansion.

The onshore wind power distribution in the *Cost-optimal* scenario shows the highest concentration in the north and leads to longer transportation ways to load centers across the country (cf. Fig. 6). This is

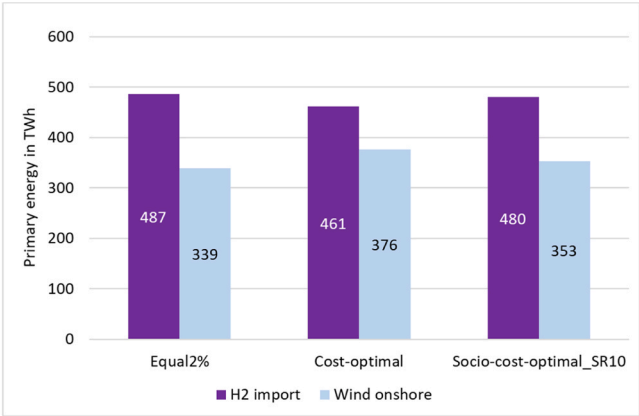


Fig. 5. Comparison of H₂ import and onshore wind energy as flexible primary energy supply per scenario.

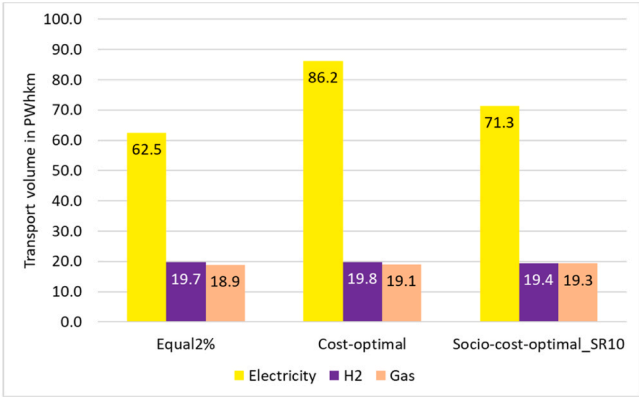


Fig. 6. Accumulated electricity energy flows for all considered branches, every time step, and weighted with the distance between centroids of regions per scenario.

further amplified by the increased electricity volume due to the efficient allocation and adds up to 38 % more electricity transport than in the *Equal2%* scenario, in which the onshore wind power capacity has higher load proximity. This is also supported by the number of self-sufficient nodes. Whereas in the *Cost-optimal* scenario, only five states produce more electricity than they consume throughout the year (including sector coupling), there are eight in the *Equal2%*. As for the aforementioned evaluation criteria, the electricity transport within the *Socio-cost-optimal_SR10* scenario lies in-between, however, besides the city-states and Saarland, only the three load centers North Rhine-Westphalia (46 %), Baden-Württemberg (80 %), and Hesse (77 %) generate less electricity than they need locally.

As we did not specify H₂ import locations and omit transit flows, the energy transport of H₂ and gas is not affected significantly by the scenarios.

3.3.4. Environmental sustainability

We measure environmental sustainability with the model-endogenous environmental costs (cf. Eq. 2), which we validate with an ex-post analysis based on conflict risks for the given capacity potential data (Guidehouse et al., 2022). As the maximum SR parameter as weight is only applied in the socio-cost-optimal scenarios, we use the same value of 10 M EUR/km² for all scenarios.

The results are provided in Table 7. The contrast of eight federal states with full and four other states without any capacity potential utilization in the *Cost-optimal* scenario allows the consideration of environmentally friendly turbine allocation only in Baden-Württemberg, Hesse, Mecklenburg-Vorpommern, and Rhineland-Palatinate. This differentiates the scenario from the others. Except for the city-states, an equal 2 % land consumption results in a capacity potential utilization for all other states between 22 % in Brandenburg (high potential) and 82 % in North Rhine-Westphalia (low potential). In the *Socio-cost-optimal_SR10* scenario, the interval is even smaller, given the narrow bounds of 40 % in Bavaria (low energy yield) and 60 % in Schleswig-Holstein (high energy yield).

To validate our model-endogenous key indicator, we compare the scenario distribution results ex-post with the provided onshore wind-eligible land area assessment of Guidehouse et al. (2022) that classifies the land area with five conflict risk values (CRV), in which CRV1 indicates a very low and CRV5 an overlapping, very high conflict risk. We assume that the land area with the lowest conflict risks is preferentially used within a state.

Fig. 7 illustrates the utilized land area by conflict risk of the different onshore wind power distributions, amounting to 7250 km² in total for each scenario. Up to 99 % of the onshore wind power capacity within the *Equal2%* scenario can be allocated to the land area with CRV3 or lower, only covering the land area in Mecklenburg-Vorpommern and the city-states with CRV4 or higher, whereas in Brandenburg land area with CRV2 remains unused. The onshore wind power capacity in the *Socio-cost-optimal_SR10* scenario is allocated to the land area with the lowest conflict risks, except for leaving 8 % of the land area in Bavaria with CRV2 unused. To realize the cost-optimal distribution, land areas of all conflict risks need to be afforded. The full utilization of land area with CRV1 can be traced back to the overlap of low conflict risks and high energy yields (Schleswig-Holstein and Lower Saxony) and high energy demand (North Rhine-Westphalia) and, therefore, can be considered a fortunate concurrence.

Table 7

Model-endogenous environmental costs per scenario for a maximum SR of 10 M EUR/km².

Scenario	Environmental costs in G EUR
Equal2%	20.088
Cost-optimal	32.614
Socio-cost-optimal_SR10	16.476

3.3.5. Distributive justice

The installation of onshore wind power capacities can be associated with both, positive and negative regional impacts, such as value creation and landscape concerns, respectively (Zerrahn, 2017; Ram et al., 2020). We evaluate how fair onshore wind power capacity is distributed between the states by applying the Gini coefficient,⁹ which indicates the inequality of distributions (1 is maximum inequality, whereas 0 means equality) (Gini, 1912). Therefore, we use the capacity potential utilization as a comparable value, which considers the different environmental realities of the regions by incorporating the individual eligible land area potential for onshore wind power capacity. To address the heterogeneity of the states in terms of population and area, we apply different weights. In the first case a), every state has the same weight, whereas in case b), the population and in case c), the land area for each state are used as weights.

Fig. 8 provides the Lorenz curves and Gini coefficients of the cases and scenarios with the potential capacity utilization index on the y-axis, in which the maximum (100) refers to the (weighted) accumulated relative potential capacity utilization for each scenario.

For the *Equal2%* scenario, the over-utilization of capacity potential within the city-states has a strong impact; especially in case a), in which all states are equally considered but also in b) due to the high population density. An exception for city-states could improve distribution equality, which can be suggested from case c) that considers their relatively small land area. The latter case, however, shows the highest inequality in the *Cost-optimal* scenario, in which states with 40 % of the whole capacity potential do not supply onshore wind energy. This emphasizes the results' dependency on the chosen metric and the associated weight behind it. Regardless of the chosen weight, the *Socio-cost-optimal_SR10* scenario provides consistently the most equal results (Gini coefficient <0.06), which is favored within the optimization problem and, consequently, tends close to an equal capacity potential utilization.

4. Discussion

4.1. Impact of onshore wind power distribution on key objectives of the energy supply

The scenarios of this study display different strategies how to distribute onshore wind power. Table 8 offers a final evaluating comparison of the impact on proposed key objectives of energy supply for the three highlighted distribution scenarios based on the impact analysis in Section 3.3.

An equal distribution of onshore wind power capacity based on the land area (represented by the *Equal2%* scenario) provides a relatively balanced electricity generation and consumption. Therefore, less energy transport is needed, which has a positive effect on system reliability and the need for grid extension (Flachsbarth et al., 2021). However, due to less efficient allocation, the reduced onshore wind energy supply increases the dependency on costly H₂ imports. Moreover, the focus on the land area does not consider the heterogeneous pre-conditions of eligible capacity potential, leading to unequal capacity potential utilization between the states. In contrast, in a cost-optimized distribution, renewable sites with mostly high energy yield are utilized, resulting in a concentration of electricity surplus in the north. This improves affordability and political independence but comes at higher costs for increased transport, negative environmental impact, and unequal distribution of regional benefits and burdens.

An onshore wind power distribution with similar capacity potential utilization (represented by the *Socio-cost-optimal_SR10* scenario) offers a compromise between equal land area consumption and efficiency. By taking into account individual capacity potential per state, it addresses

⁹ Defined as the area between the line of equality and the corresponding scenario line (Lorenz curve) multiplied by two.

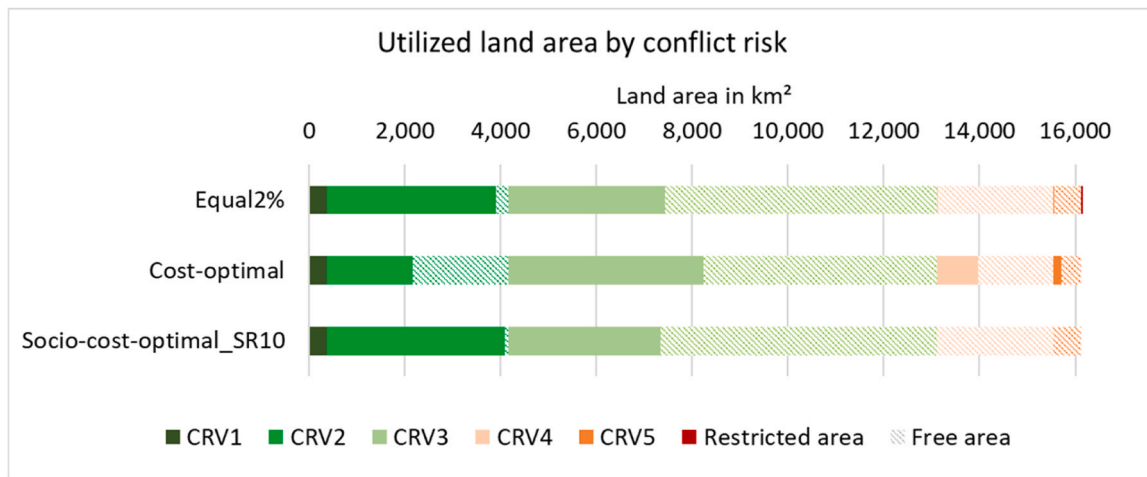


Fig. 7. Utilized land area by conflict risk value indicated by colored bar per scenario.

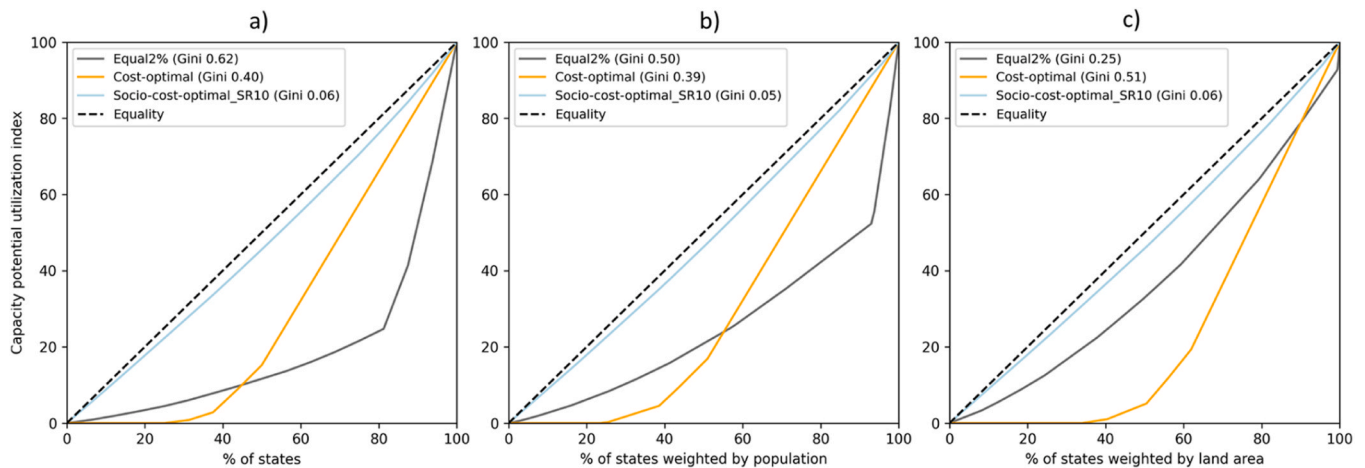


Fig. 8. Lorenz curves and Gini coefficients for scenarios indicating the inequality of capacity potential utilization of the a) states, b) states weighted by population, and c) states weighted by land area.

Table 8

Overview of relative impact on energy system evaluation criteria and indicators per wind energy distribution scenario.

Criteria	Indicators	Equal2%	Cost-optimal	Socio-cost-optimal_SR10
Affordability	Total system costs	+2.5 G EUR		+1.8 G EUR
Energy independence	H ₂ imports	+5.6%		+4.1%
System reliability	Electricity transport		+38%	+14%
Environmental sustainability	Environmental costs	+22%	+98%	
Distributive justice	Equality	Gini: 0.25	Gini: 0.51	Gini: 0.06

the geological differences of the considered regions, allowing land areas with higher conflict risks to be spared. Besides the small economic weight of onshore wind power capacities, also the correlation of eligible capacity potential and wind sites favors the socio-cost-optimization compared to the equal land consumption scenario.

Throughout all onshore wind power distribution scenarios, we observe a similar energy system structure. All components in the model are utilized, with the highest sensitivity of electrolysis capacity (+9 % based on *Equal2%* scenario) and H₂ import compensation for inefficient onshore wind energy allocation (+5 %). Consequently, the total energy system costs differ moderately, by up to 2.4 G EUR annually. In contrast,

we find electricity transport to be sensitive to onshore wind allocation (+38 %). Our ex-post analysis of land consumption suggests distribution results influence environmental sustainability by requiring land area with varying conflict risks. Ultimately, despite individual inequality metrics being arguable, the presented scenarios show clear differences regarding their associated distributive justice.

4.2. Contribution of energy system analysis to onshore wind power distribution

The utilization of an energy system optimization model allows

considering the spatio-temporal value of the onshore wind energy as a part of the energy supply linked to energy demand. In general, the consideration of transport, storage, and sector-coupling ensures high technical feasibility and enhances the transparency for the subsequent impact on the energy system and its components. This way, economic integration can be achieved, which reduces curtailment and total energy system costs. To account for the different objectives of the scenarios proposed in this study, we compare the resulting onshore wind power distributions with different approaches in the literature.

Similar to the results of cost-optimized spatially-explicit turbine placement (Tafarte and Lehmann, 2023; Weinand et al., 2022b), the main driver of cost-optimized onshore wind power allocation within an energy system optimization model is the local energy yield. However, since limited transport capacities and transport losses are included, the cost-optimal onshore wind power distribution does not strictly favor the sites with the highest yields, as illustrated in Fig. 9. Particularly in the northeast, the capacity potential remains unused. In addition to the high load centers allocated in the west and south, the supply of offshore wind energy in Mecklenburg-Vorpommern provides already three times the local electricity consumption, which is partly compensated by flexible local H₂ production. For Brandenburg and Saxony-Anhalt, all neighboring states have higher capacity factors on average, and even capacity potential with higher yield is left out. While load proximity has also been mentioned as an equity criterion pursuing commutative justice (Lehmann et al., 2021), the results reflect the systemic value of load proximity, which to consider improves economic viability. However, the appropriateness of the final concentrated allocation remains questionable. In contrast to other studies, the model's lower spatial resolution and applied data neglect existing heterogeneous intraregional yield distributions (Masurowski, 2016), including outliers, and explicit turbine siting is omitted.

For high space resistance parameter values in our socio-cost-optimal scenarios, the onshore wind power distribution tends to equal utilization of capacity potential and, thus, addresses an equity factor similar to equal land consumption. In our *Socio-cost-optimal_SR10* scenario, the average environmental costs (46.7 EUR/MWh) clearly exceed the suggested generation costs (33.1 EUR/MWh). As energy yield and system integration matter less for the allocation itself (or in the *Equal2%* scenario not at all), the application of an energy system model is less needed to arrive at the resulting capacity distributions. These are mainly determined by the input data, such as the capacity potential, and a chosen equity factor.

Fig. 10 compares the three highlighted scenarios of this study regarding their land consumption per state with the approaches by

Guidehouse et al. (2022) and Stiftung Klimaneutralität (2021), who also include equity aspects in their capacity distributions. Although the applied methods and data for the allocation differ (except the same capacity potential in this study and Guidehouse et al. (2022)), the resulting distributions show similarities to the *Socio-cost-optimal_SR10* scenario, while also tending slightly toward equal land consumption of 2 %. The highest deviations from our scenario can be observed in Baden-Württemberg with relatively low potential and yield (+1.3 p.p.) and in Brandenburg with the highest potential (-1.4 p.p.), which also exceeds the values in comparable other potential studies clearly. The average non-generation costs (environmental costs) of our scenario similar to the two selected distributions from the literature are well above the upper disamenity costs results for Germany of Ruhnau et al. (2022). This suggests implicitly either a higher weighting of avoiding disamenities or the consideration of more (additional) objectives compared to our cost function by the authors.

5. Conclusion

In this study, we widened the question of how to allocate onshore wind power distribution from an energy system perspective for the case of climate-neutral Germany. Thereby, we wanted to understand, which influence the distribution of onshore wind power has on key objectives of the energy supply. Furthermore, we explored the potential contribution energy system analysis as a discipline can provide to support decision-making regarding onshore wind power distribution.

First, we find the impact of onshore wind power distribution on the energy system to be moderate, which eases consideration of further criteria and is in agreement with diverse previous research (Neumann and Brown, 2021; Lehmann et al., 2023). Throughout all distribution scenarios that were considered, the general energy system structure remained insensitive, comprising the same components as storage and sector-coupling in similar quantities. The most sensitive components in terms of quantity and distribution are electrolyzers, which serve as flexible electricity consumers and are generally allocated at sites with electricity surpluses. While we assumed a fully integrated energy system with cooperating sector-coupling and multi-modal transport providing system flexibility, we neglected demand-side-management (short-term) and load allocation (long-term), which can react to price signals and may even increase the system design reactivity to different onshore wind power distributions. Furthermore, we observe relatively low reductions of annual system costs by 2.4 G EUR and 5 % of H₂ imports for a cost-optimal distribution scenario compared to a distribution with equal onshore wind power land consumption per state. On the other hand, there is a trade-off as this comes with an increase of 38 % for electricity transport due to more electricity generation and concentrated onshore wind energy sites. A more amplified approach with broader spatial coverage and higher resolution as well as more constrained H₂ imports and potential grid expansion might shift this observation and could be tested in the future.

Besides the differences in techno-economic indicators, the scenarios show disparate equality of the utilization of onshore wind power capacity potential. However, taking into account the heterogeneous characteristics of the states regarding the eligible land area, also an equal land consumption per state is not necessarily equitable as states with relatively lower capacity potential may be impacted significantly higher. Our results yield diverging conclusions regarding the equality of the distribution scenarios depending on chosen weights, which is why further debates about apt equity metrics are necessary. This becomes increasingly important, as in contrast to historic onshore wind power siting, which complied with economic principles (Weinand et al., 2022a), current suggestions for energy systems with high renewable energy share include and emphasize equality for onshore wind power distribution.

Second, we conclude that energy system optimization can provide its own solutions for onshore wind power distributions, increase

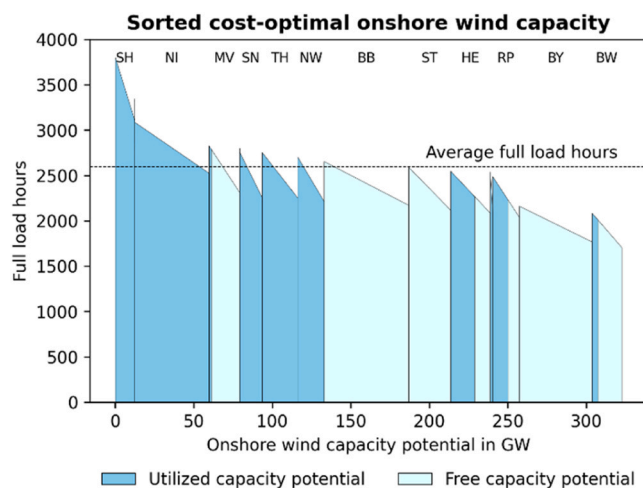


Fig. 9. Onshore wind power capacity potential sorted by full load hours per federal state indicating, which capacity potential is utilized in the Cost-optimal scenario.

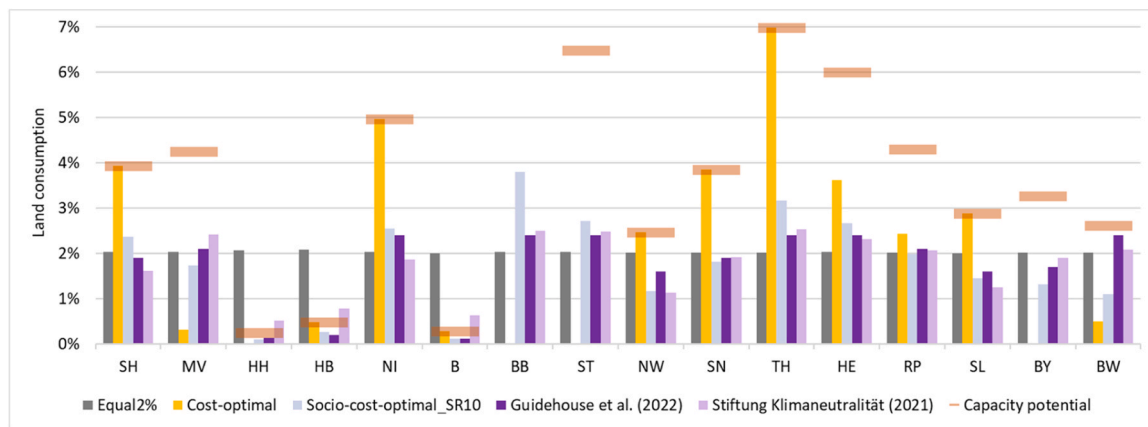


Fig. 10. Comparison of land consumption per federal state by onshore wind power distribution scenarios with selected literature.

transparency regarding trade-offs between different distributions, and, thereby, also improve other distribution methods that do not necessarily rely on optimization.

Our socio-cost-optimal approach focuses on the capacity potential and favors equal utilization while still considering energy yield and energy system integration costs. In contrast to the cost-optimal scenario, it shows similarities to results in current literature with a range of capacity potential utilization between 40 % and 60 % in its outermost scenario and highlights relevant criteria such as environmental sustainability and distributive justice. Even though the presented approach requires further research effort regarding the calibration and the verification for more cases, the inclusion of multi-criteria should be adopted in spatially-distributed energy system analysis (Flachsbarth et al., 2021; Süsser et al., 2022) for all scenarios, in which renewable energy is dominating and not just marginally complementing. Besides the support for actual siting decisions, it can serve as a better approximation than cost-optimal siting, which can be helpful with related topics, such as creating grid expansion scenarios.

For given onshore wind power distributions, energy system analysis allows examining the techno-economic feasibility by linking the spatio-temporal feed-ins with projected energy consumption in detailed bottom-up models. Therefore, results can also contain information about the systemic consequences, such as the need for grid expansion, storage, or other flexibilities, and their impact on the general goals of the energy supply. This increases transparency as it permits the quantification of costs for an end-use and relevant trade-offs, for instance between efficient allocation and exploiting other energy sources. Whereas the chosen energy system evaluation criteria in this study cover essential objectives of energy policy, the associated key indicators, in particular for system reliability, environmental sustainability, and distributive justice, still leave room for improvement and discussion.

Besides the optimization of onshore wind power distributions and their assessment, insights and conclusions from energy system analysis can be integrated into non-optimization methods, which typically concentrate on sophisticated data for capacity potential and energy

yield, need fewer resources, and can be interpreted better. Even though, our results demonstrate the high flexibility potential of the energy system, for more cost-optimal-oriented onshore wind distributions, other renewable energy supply and (a minimum) load proximity should be considered to not overestimate transport capacities. This becomes increasingly important in scenarios with more end energy electrification when sector-coupling and transport via the gas and H₂ infrastructure become less economic. As we applied only one end energy scenario, the validity range should be specified in future research by analyzing trade-offs of onshore wind power siting, grid expansion, and end energy technologies in heat, mobility, and industry sectors. Lastly, we also preset all renewable capacities and neglected interdependencies with solar energy allocation, which remains a challenge to explore in future research.

Funding

This work was supported by the Ministry of Science and Culture of Lower Saxony (Niedersächsisches Ministerium für Wissenschaft und Kultur (MWK))(grant no VWZN3770 and 74ZN1596).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

None

Appendix A

Nomenclature

Table A9
Model notation – Sets.

Symbol	Description
$b \in B$	Set of branches
B_o	Subset of branches in cycle o
$c \in C$	Set of components
$e \in E$	Set of energy carriers
$k \in K$	Set of nodes
$o \in O$	Set of cycles
$t \in T$	Set of time steps
$x \in X$	Set of converters
X^{dual}	Subset of converters with dual input
X^{ext}	Subset of converters with extraction combined heat and power

Table A10
Model notation – Parameters

Symbol	Unit	Description
β_c^{rf}		Capital recovery factor of component c
c_e^{car}	EUR/GWh	Specific fuel cost of energy carrier e
c_c^{fix}	EUR/GW	Specific fix costs for capacity of component c
c_c^{om}	/ c_c^{fix}	Specific operation and maintenance costs for capacity of component c
$c_{c,\text{max}}^{\text{sr}}$	EUR/km ²	Maximum space resistance against land consumption
ϑ	GW/km ²	Capacity density of onshore wind power
$p_{c,k}^{\text{max}}$	GW	Capacity potential of component c at node k
η_c		Efficiency rate of component c
σ_x^{cv}		Power loss factor of chp converter x
σ_x^{ch}		Minimum power-to-heat factor of chp converter x
x_b	Ω	Reactance of branch b
$\kappa_{b,o}^{\text{dir}}$		Direction of branch b in cycle o

Table A11
Model notation – Variables

Symbol	Unit	Description
C^{sys}	EUR	Annual energy system costs
C^{env}	EUR	Annual environmental costs
$E_{e,k}^{\text{import}}$	GWh	Imported energy of energy carrier e at node k
$p_{c,k}^{\text{cap}}$	GW	Power capacity of component c at node k
$p_{c,k,t}^{\text{out}}$	GW	Power output of component c at node k in time step t
$p_{c,k,t}^{\text{in}}$	GW	Power input of component c at node k in time step t
$p_{b,t}^{\text{flow1}}$	GW	Power flow (+) in branch b in time step t
$p_{b,t}^{\text{flow2}}$	GW	Power flow (-) in branch b in time step t

Equations

DC power flow model:

The sum of all power flows $p_{b,t}^{\text{flow}}$ over all branches in a cycle o multiplied by the reactance x_b must be zero. To account for the given orientation of a branch b in the cycle, the power flows are directed by the cycle matrix $\kappa_{b,o}^{\text{dir}}$.

$$\sum_{b \in B_o} (p_{b,t}^{\text{flow1}} - p_{b,t}^{\text{flow2}}) \cdot \kappa_{b,o}^{\text{dir}} \cdot x_b = 0 \forall o, \forall t \quad (\text{A3})$$

Dual converter input:

We group converters x , which can have two interchangeable input energy carriers, to the set of dual input converters X^{dual} . In the present model, H₂ and gas (methane) can both be used for the power input $p_{x,k,t}^{\text{in}}$ of a combined cycle power plant in any time step t and node k .

$$p_{x,k,t}^{\text{in}} = p_{x,k,t}^{\text{in},1} + p_{x,k,t}^{\text{in},2} \forall x \in X^{\text{dual}}, \forall k, \forall t \quad (\text{A4})$$

Combined heat and power model with extraction turbine:

The power input $p_{x,k,t}^{\text{in}}$ of combined heat and power plants X^{ext} can be calculated with primary $p_{x,k,t}^{\text{out}}$ and secondary $p_{x,k,t}^{\text{out,th}}$ (thermal) power output,

power loss factor σ_x^{cv} and efficiency rate η_x . Furthermore, the installed power capacity $P_{x,k}^{cap}$ must not be exceeded.

$$P_{x,k,t}^{out} + P_{x,k,t}^{out,th} \cdot \sigma_x^{cv} = P_{x,k,t}^{in} \cdot \eta_x \leq P_{x,k}^{cap} \forall x \in X^{ext}, \forall k, \forall t \quad (A5-6)$$

For a secure operation, a minimum power-to-heat factor σ_x^{cb} must be considered.

$$P_{x,k,t}^{out} \geq P_{x,k,t}^{out,th} \cdot \sigma_x^{cb} \forall x \in X^{ext}, \forall k, \forall t \quad (A7)$$

Tables

Table A12

Annual end energy per energy carrier specified by region.

	End energy in TWh				
	Electricity	District Heating	H ₂ / Gas	Oil	Biomass
BW	79	16	23	8	8
BY	100	21	31	32	10
B	15	5	3	3	3
BB	24	4	10	2	2
HB	9	1	6	1	1
HH	13	3	3	7	1
HE	46	10	13	35	5
MV	10	2	3	0	1
NI	69	14	26	2	7
NW	166	28	75	135	14
RP	37	7	13	33	3
SL	18	2	12	0	1
SN	27	6	9	22	3
ST	27	4	14	14	2
SH	17	5	5	7	2
TH	16	3	5	1	2
DE	672	133	259	301	68

Table A13

Maximum annual total electricity generation for solar, offshore wind, and hydro power plants specified by region

	Maximum total electricity generation in TWh		
	Solar	Offshore wind	Hydro
BW	44	-	3.8
BY	89	-	8.6
B	0.7	-	0
BB	23	-	0
HB	0.2	-	0.1
HH	0.3	-	0
HE	20	-	0.3
MV	19	26	0
NI	43	143	0.2
NW	44	-	0.7
RP	18	-	0.9
SL	3	-	0
SN	17	-	1.0
ST	20	-	0.4
SH	15	81	0
TH	14	-	0.7
DE	370	249	16.6

Table A14

Overview of costs and other parameters of technologies. To derive yearly capital expenditures for the installation of technologies, we apply weighted average costs of capital of 2.5 % to calculate the capital recovery factor.

Technologies	Specific fix costs		O&M costs	Lifetime a	Efficiency	Decay %/day	Power loss factor	Power/ Heat factor
	M EUR/GW _{out}	M EUR/GWh						
Onshore wind power	900		3.5 %	25				
Electrolysis	360		5.0 %	20	0.67			
Large-scale heat pump	490		0.4 %	25	2.90*			

(continued on next page)

Table A14 (continued)

Technologies	Specific fix costs		O&M costs	Lifetime a	Efficiency	Decay %/day	Power loss factor	Power/ Heat factor
	M EUR/GW _{out}	M EUR/GWh						
Combined cycle power plant	963		4.0 %	30	0.56		0.17	0.42
Biomass power plant	2900		4.5 %	30	0.38		0.15	1.20
Biogas upgrader	1600		5.1 %	25	0.70			
Methanation	235		5.0 %	20	0.70			
Fischer Tropsch	235		5.0 %	20	0.70			
Battery storage	30	130	1.0 %	15	0.95	0.12		
Pumped storage	-	-	-		0.86	0		
H ₂ storage	575	0.45	2.5 %	30	0.93	0		
Gas storage	-	-	-		0.93	0		
Thermal storage	0.00	10	0.5 %	25	0.90	0.72		

^a average value for the coefficient of performance that depends on local temperature

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